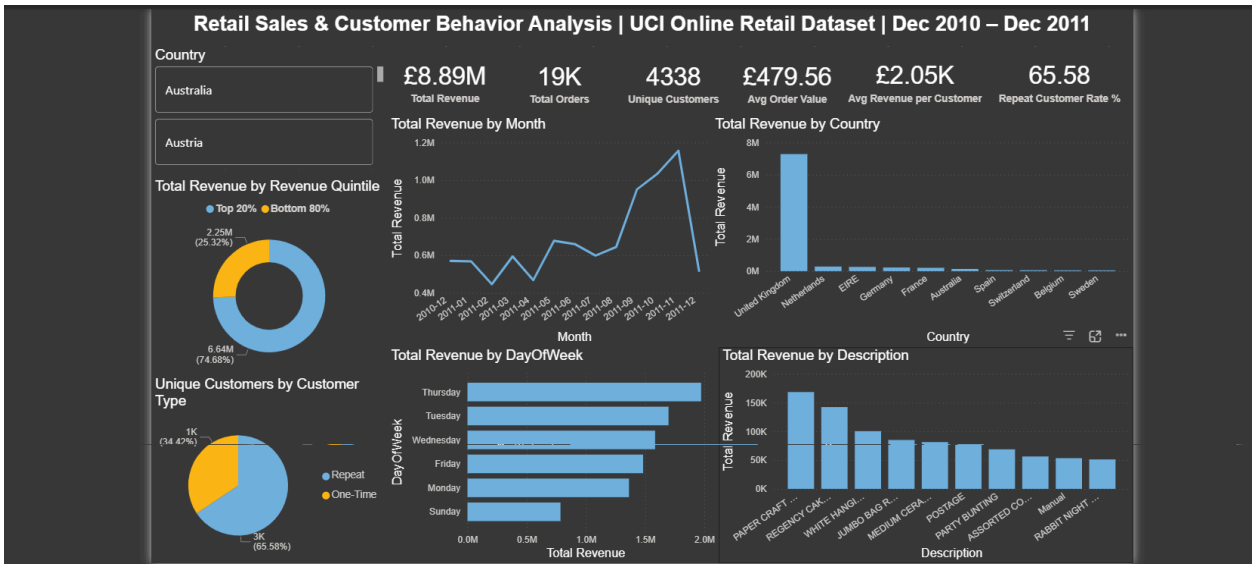
Retail Sales & Customer Behavior Analysis
Executive Report
Dataset: UCI Online Retail Dataset
Analysis Period: December 2010 to December 2011



1. Executive Summary

This analysis combines findings from the Jupyter notebook, SQL exploration, and Power BI dashboard to evaluate sales performance, customer behavior, product demand, seasonality, and geographic revenue concentration. After cleaning the raw transactional dataset, the final analytical dataset contained 392,692 valid sales records. Across this period, the business generated total revenue of approximately GBP 8.89 million from 18,532 orders and 4,338 unique customers.

The analysis shows that revenue performance was strong, but highly concentrated. The United Kingdom accounted for the vast majority of total revenue, while a relatively small group of high-value customers drove most of the business. Customer repeat behavior was meaningful, with more than 65% of customers placing more than one order. Product-level analysis also showed that a limited number of items generated a disproportionate share of both sales volume and revenue.

Taken together, the notebook, SQL layer, and dashboard tell a consistent story: the business has strong core demand, a loyal repeat customer base, clear seasonal peaks, and high dependence on

both top customers and the UK market. These findings are valuable for retention planning, market expansion, inventory prioritization, and executive reporting.

2. Data Preparation and Quality Review

The original dataset contained 541,909 transaction rows. Before analysis, the notebook performed a structured data cleaning process to remove records that would distort business insights. Key issues identified in the raw data included missing customer identifiers, duplicate rows, cancelled invoices, negative quantities, and zero or negative unit prices.

The major data quality findings were as follows:

Missing CustomerID rows: 135,080
Duplicate rows: 5,268
Cancelled invoices: 9,288
Negative quantity rows: 10,624
Zero or negative unit price rows: 2,517
To create an analysis-ready dataset, the notebook removed rows with missing customer IDs, exact duplicates, cancelled invoices, and invalid sales values. After cleaning, the dataset was reduced to 392,692 valid rows. All remaining records represented completed purchases with positive quantity and positive unit price, making the dataset suitable for KPI reporting, customer segmentation, product analysis, and dashboarding.

Data Cleaning

The cleaning steps below create an analysis-ready dataset focused on valid completed purchases. The rules are intentionally simple and business-friendly:

- remove rows with missing `CustomerID` for customer-level analysis
- remove exact duplicates
- remove cancelled invoices
- keep only positive quantities and positive prices
- standardize data types for dates and customer IDs

```python
df = df.dropna(subset=['CustomerID']).drop_duplicates().copy()
df = df[~df['InvoiceNo'].astype(str).str.startswith('C')].copy()
df = df[(df['Quantity'] > 0) & (df['UnitPrice'] > 0)].copy()

df['CustomerID'] = df['CustomerID'].astype(int)
df['InvoiceDate'] = pd.to_datetime(df['InvoiceDate'])

print(f"Cleaned dataset shape: {df.shape}")
print(f"Remaining null values: {df.isnull().sum().sum()}")
print(f"Remaining duplicates: {df.duplicated().sum()}")
```

[10] Python

```
Cleaned dataset shape: (392692, 8)
Remaining null values: 0
Remaining duplicates: 0
```

```python
post_clean_checks = pd.Series({
    'rows_after_cleaning': len(df),
    'remaining_null_customerid': int(df['CustomerID'].isna().sum()),
    'remaining_duplicates': int(df.duplicated().sum()),
    'remaining_non_positive_quantity': int((df['Quantity'] <= 0).sum()),
    'remaining_non_positive_price': int((df['UnitPrice'] <= 0).sum()),
    'date_range_start': df['InvoiceDate'].min(),
    'date_range_end': df['InvoiceDate'].max()
})
post_clean_checks
```

[11] Python

```
rows_after_cleaning                          392692
remaining_null_customerid                         0
remaining_duplicates                              0
remaining_non_positive_quantity                   0
remaining_non_positive_price                      0
date_range_start                  2010-12-01 08:26:00
date_range_end                    2011-12-09 12:50:00
dtype: object
```

The validation output confirms that the final analytical dataset no longer contains the key issues targeted in the cleaning step.

3. KPI Snapshot

The cleaned dataset produced the following headline business metrics:

Total Revenue: GBP 8,887,208.89
Total Orders: 18,532
Unique Customers: 4,338
Unique Products: 3,665
Average Order Value: GBP 479.56
Average Revenue per Customer: GBP 2,048.69
Repeat Customer Rate: 65.58%
These KPIs indicate a business with relatively strong order values and healthy customer monetization. Average order value near GBP 480 suggests that transactions are not low-ticket purchases alone, while average revenue per customer above GBP 2,000 indicates meaningful customer lifetime value within the observed period.

£8.89M	19K	4338	£479.56	£2.05K	65.58
Total Revenue	Total Orders	Unique Customers	Avg Order Value	Avg Revenue per Customer	Repeat Customer Rate %

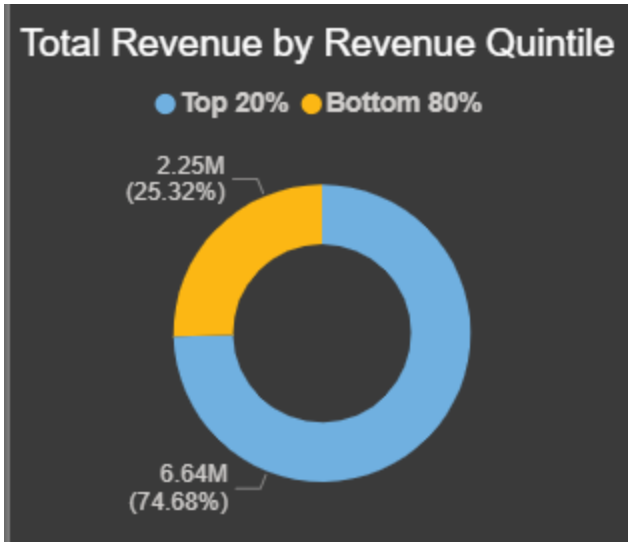
4. Customer Behavior Analysis

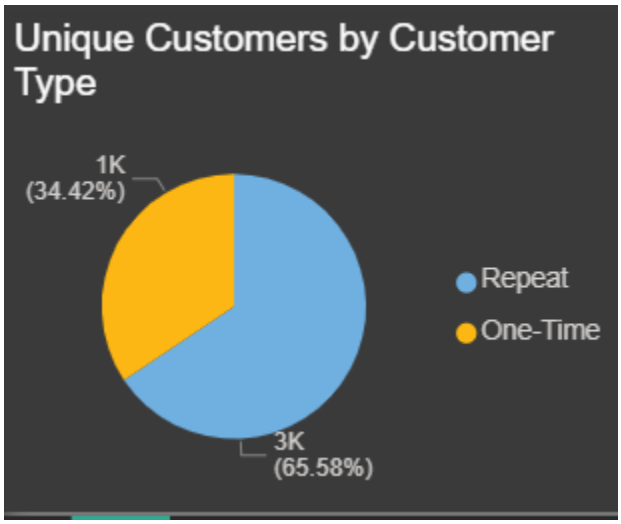
Customer behavior analysis revealed that repeat purchasing is a major component of business performance. Out of 4,338 customers, 2,845 placed more than one order, which translates to a repeat customer rate of 65.58%. This suggests that the business is not driven solely by one-time transactions and benefits from measurable customer retention.

A deeper segmentation analysis showed a strong Pareto pattern. The top 20% of customers, equal to 868 customers, generated 74.68% of total revenue. This means the majority of revenue came from a relatively small group of high-value customers. The pattern became even more important when repeat behavior was examined: top-revenue customers were responsible for 59.73% of repeat orders. This confirms that the most valuable customers are not only high spenders, but also more engaged and more likely to come back.

The highest-value individual customers included:

Customer 14646: GBP 280,206.02 from 73 orders
Customer 18102: GBP 259,657.30 from 60 orders
Customer 17450: GBP 194,390.79 from 46 orders
Customer 16446: GBP 168,472.50 from 2 orders
Customer 14911: GBP 143,711.17 from 201 orders
These results support targeted retention and account management strategies. High-value customers should be prioritized for loyalty programs, personalized offers, premium service treatment, and churn prevention initiatives.





5. Product Performance Analysis

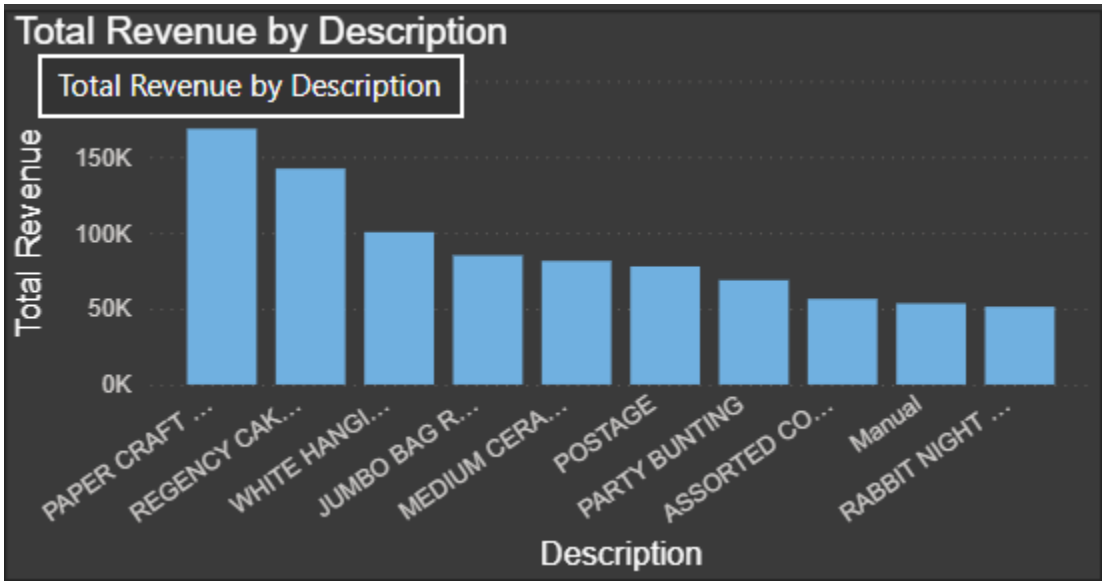
Product-level analysis identified the top-performing items by both revenue and quantity sold. The highest revenue-generating product was PAPER CRAFT , LITTLE BIRDIE, which generated GBP 168,469.60. Other top revenue contributors included REGENCY CAKESTAND 3 TIER, WHITE HANGING HEART T-LIGHT HOLDER, JUMBO BAG RED RETROSPOT, and MEDIUM CERAMIC TOP STORAGE JAR.

Top products by revenue were:

PAPER CRAFT , LITTLE BIRDIE: GBP 168,469.60
REGENCY CAKESTAND 3 TIER: GBP 142,264.75
WHITE HANGING HEART T-LIGHT HOLDER: GBP 100,392.10
JUMBO BAG RED RETROSPOT: GBP 85,040.54
MEDIUM CERAMIC TOP STORAGE JAR: GBP 81,416.73

Top products by quantity sold were:

PAPER CRAFT , LITTLE BIRDIE: 80,995 units
MEDIUM CERAMIC TOP STORAGE JAR: 77,916 units
WORLD WAR 2 GLIDERS ASSTD DESIGNS: 54,319 units
JUMBO BAG RED RETROSPOT: 46,078 units
WHITE HANGING HEART T-LIGHT HOLDER: 36,706 units
A key takeaway is that some products appear in both the top-revenue and top-volume rankings. This indicates a combination of strong customer demand and strong commercial contribution. These products should be monitored carefully for stock availability, replenishment planning, supplier reliability, and promotional timing.



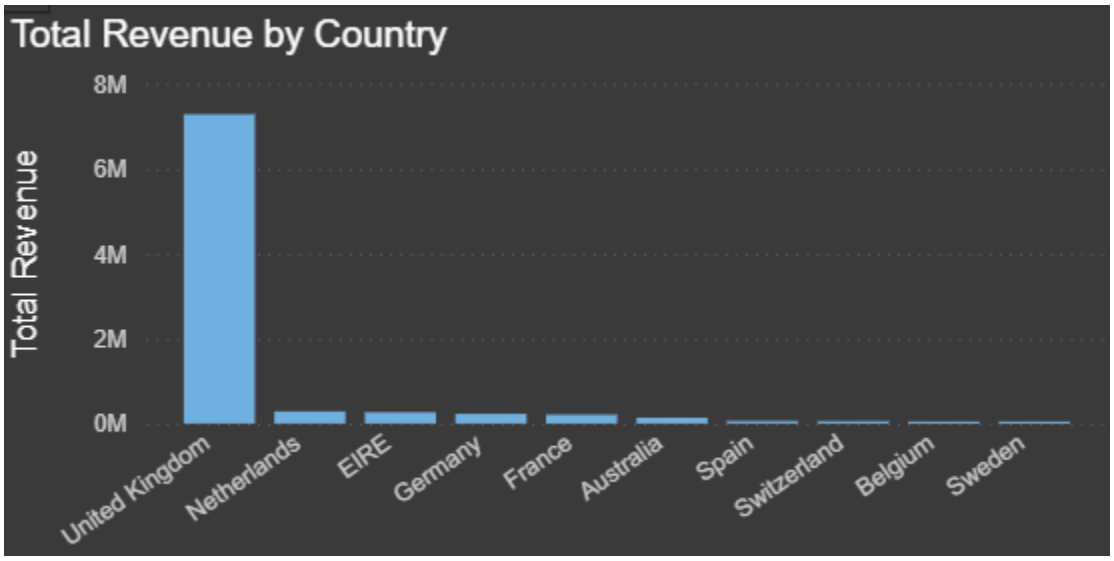
6. Geographic Revenue Analysis

Country-level analysis showed that revenue was heavily concentrated in the United Kingdom. The UK generated GBP 7,285,024.64, accounting for 81.97% of total revenue. This far exceeded all other countries and clearly established the UK as the primary revenue market.

The top revenue-generating countries were:

United Kingdom: GBP 7,285,024.64
Netherlands: GBP 285,446.34
EIRE: GBP 265,262.46
Germany: GBP 228,678.40
France: GBP 208,934.31
Australia: GBP 138,453.81

Spain: GBP 61,558.56
Switzerland: GBP 56,443.95
Belgium: GBP 41,196.34
Sweden: GBP 38,367.83
This geographic profile presents both strength and risk. The business clearly has strong traction in the UK, but such a high level of dependence on a single market creates concentration risk. Any decline in UK demand, pricing pressure, logistics disruption, or customer churn could materially impact total revenue. At the same time, existing demand in the Netherlands, EIRE, Germany, and France suggests realistic expansion opportunities in markets where the business already has a foothold.



7. Revenue Trend and Seasonality Analysis

Monthly revenue analysis revealed both volatility and strong seasonality. Revenue started at GBP 570,422.73 in December 2010, dipped in early 2011, then recovered through the middle of the year before accelerating sharply in the final quarter.

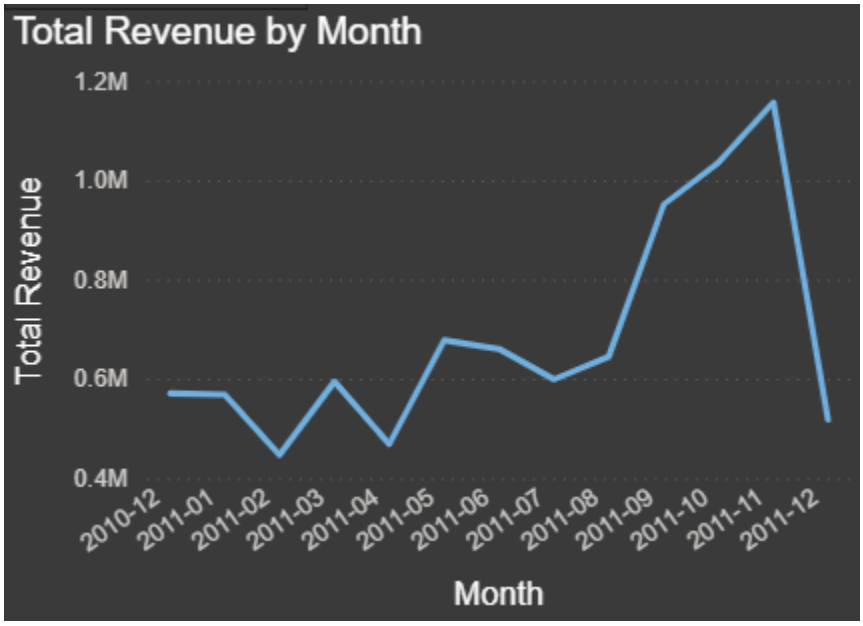
Monthly revenue values were:

2010-12: GBP 570,422.73
2011-01: GBP 568,101.31
2011-02: GBP 446,084.92
2011-03: GBP 594,081.76
2011-04: GBP 468,374.33
2011-05: GBP 677,355.15
2011-06: GBP 660,046.05
2011-07: GBP 598,962.90

2011-08: GBP 644,051.04
2011-09: GBP 950,690.20
2011-10: GBP 1,035,642.45
2011-11: GBP 1,156,205.61
2011-12: GBP 517,190.44
The strongest month was November 2011, when revenue exceeded GBP 1.15 million. September through November showed a clear surge, likely reflecting seasonal retail uplift. December 2011 shows a sharp decline, but this should be interpreted carefully because it is a partial month in the dataset and not directly comparable with earlier full months.

Month-over-month growth also highlighted major swings. Revenue fell more than 21% in February 2011, then rebounded by more than 33% in March and more than 44% in May. The strongest late-year increase came in September 2011, when revenue jumped by approximately 47.61% compared with August.

This time-series pattern indicates that the business experiences meaningful seasonality, especially in the run-up to year-end. Forecasting, staffing, inventory planning, and promotional scheduling should account for this late-year surge.



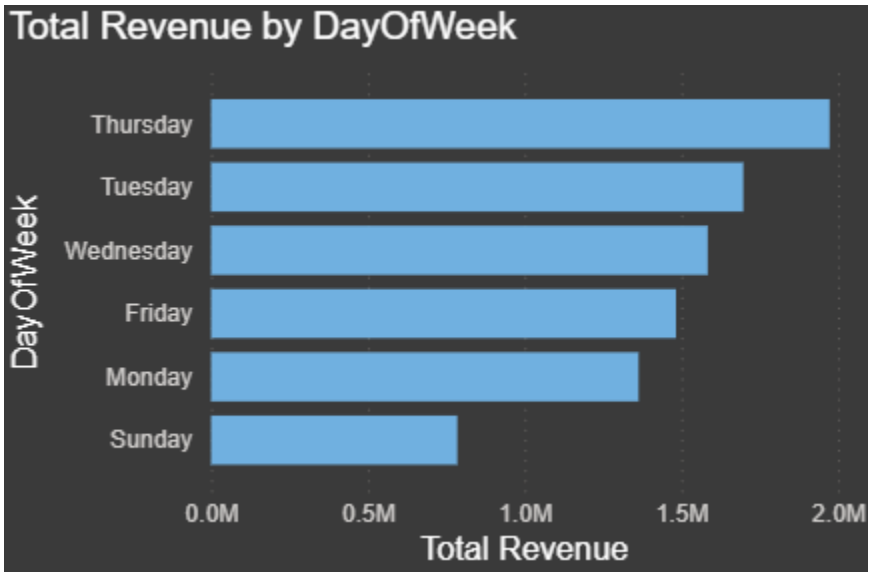
8. Day-of-Week Performance

The dashboard's day-of-week view showed that revenue was not evenly distributed throughout the week. Thursday was the highest-performing day, generating approximately GBP 1.97 million in revenue. Tuesday and Wednesday also performed strongly, while Sunday was the lowest

active trading day. No Saturday revenue appeared in the cleaned dataset, indicating either no recorded transactions or no valid completed sales for that day.

Revenue by day of week was approximately:

Thursday: GBP 1.97M
Tuesday: GBP 1.70M
Wednesday: GBP 1.58M
Friday: GBP 1.48M
Monday: GBP 1.36M
Sunday: GBP 0.79M
This pattern suggests that customer purchasing activity peaks during the business week rather than on weekends. Operationally, this can inform staffing, campaign timing, service capacity, and fulfillment planning.



9. SQL Analysis Contribution

The SQL script extends the notebook analysis into a reusable business intelligence layer. It includes structured queries for executive KPI reporting, product rankings, market contribution analysis, order value analysis, repeat purchase behavior, customer segmentation, and cohort-style customer lifetime review.

Several SQL features strengthened the analysis:

Aggregate queries created standardized executive KPI outputs.
Window functions such as RANK() were used to identify best-selling products by country.
LAG() supported month-over-month revenue growth analysis.
NTILE(5) enabled customer quintile segmentation for Pareto-style revenue analysis.

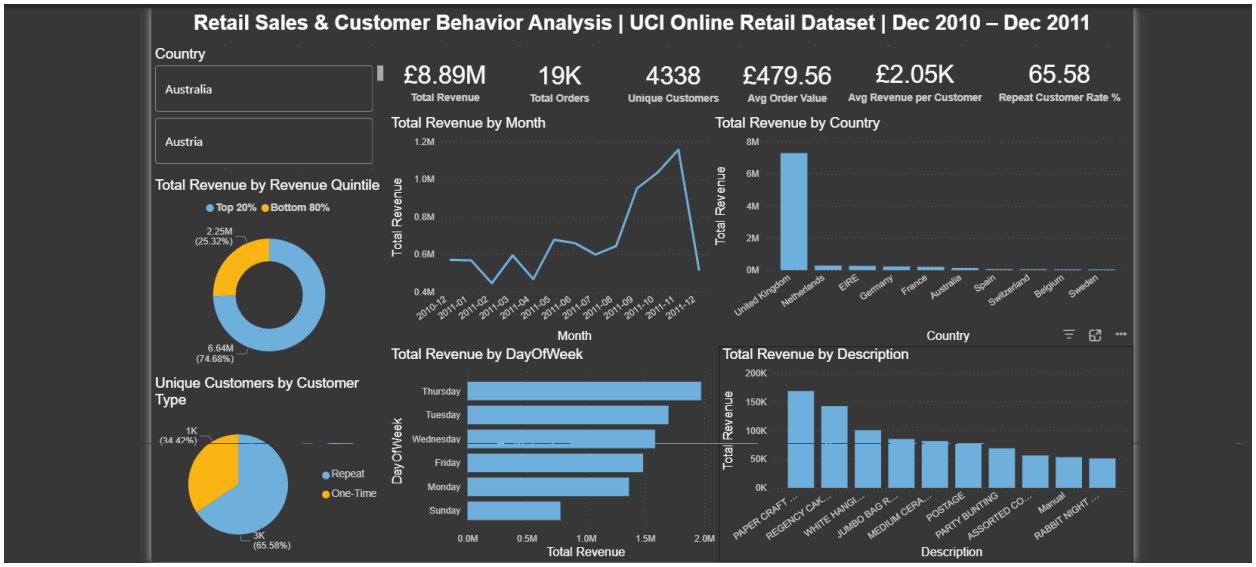
Common Table Expressions improved readability and ensured metrics like average order value were calculated correctly.
This SQL layer is important because it makes the project more production-ready. Instead of relying only on Python notebooks, the cleaned invoice table can support dashboard refreshes and stakeholder reporting directly from a database environment.

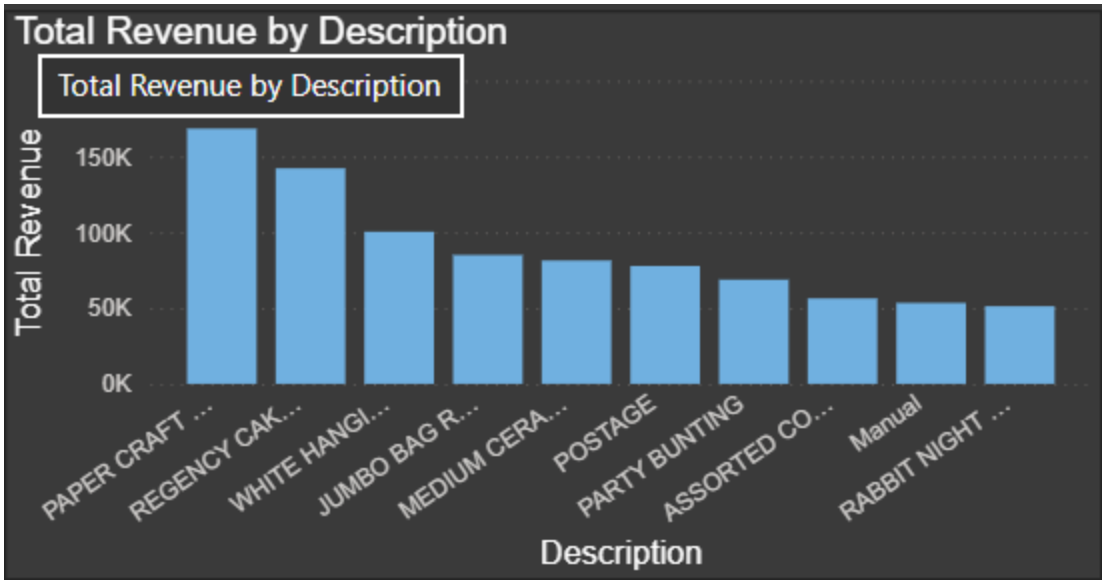
10. Power BI Dashboard Summary

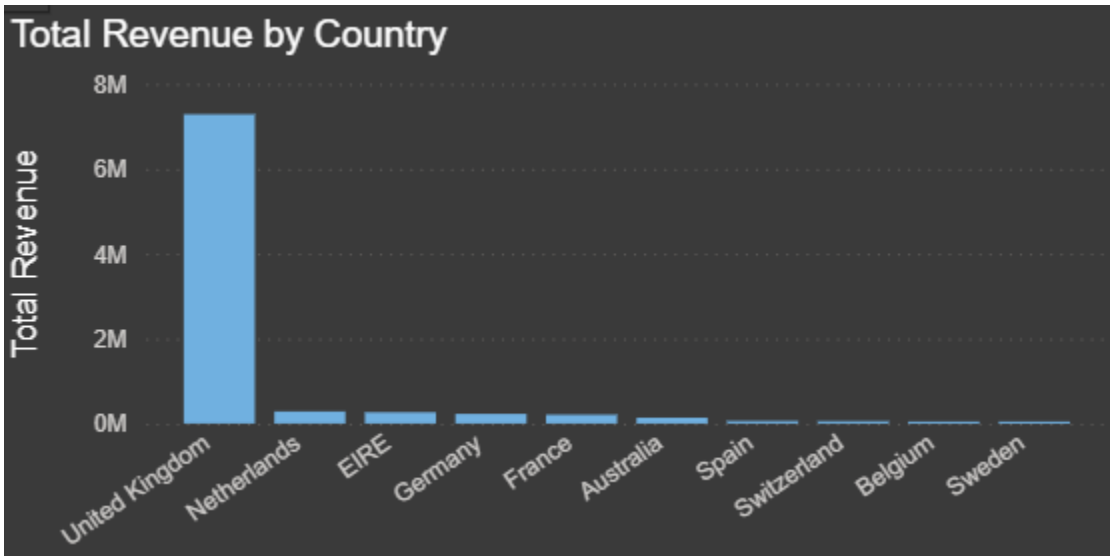
The Power BI dashboard successfully consolidates the analysis into a stakeholder-friendly executive view. It highlights the most important KPIs at the top, then supports them with charts for monthly revenue, country contribution, customer concentration, day-of-week performance, and product-level revenue.

The dashboard confirms several core findings:

Revenue is concentrated among top customers.
The UK dominates geographic performance.
Q4 2011 showed the strongest revenue momentum.
Repeat customers form the majority of the customer base.
A small group of products contributes disproportionately to business performance.
From a reporting perspective, the dashboard is effective because it turns notebook and SQL outputs into a clear visual story that can be reviewed quickly by business users, recruiters, or decision-makers.







11. Key Business Insights

The most important findings from the full analysis are summarized below:

The business generated GBP 8.89 million in revenue from 18,532 orders.
Customer monetization was strong, with average order value of GBP 479.56.
Repeat behavior was significant, with 65.58% of customers placing more than one order.
Revenue concentration was very high, with the top 20% of customers generating 74.68% of total revenue.
The same high-value segment drove 59.73% of repeat orders.
The UK contributed 81.97% of total revenue, showing heavy geographic concentration.
Revenue peaked in November 2011, confirming strong year-end seasonality.

Top products contributed strongly across both revenue and unit volume, making them critical for inventory and promotion planning.

12. Recommendations

Based on the combined findings from the notebook, SQL, and dashboard, the following actions are recommended:

Prioritize retention of top-value customers through personalized engagement, loyalty programs, and proactive account management.
Reduce dependency on the UK market by strengthening growth strategies in the Netherlands, EIRE, Germany, and France.
Protect stock availability for the highest-performing products, especially before seasonal demand peaks.
Use monthly and day-of-week demand patterns to improve staffing, marketing timing, and inventory allocation.
Monitor concentration risk regularly so the business is not overly dependent on a small customer segment or single geographic market.
Continue using the cleaned dataset and SQL reporting layer as the source for future dashboard refreshes and executive reporting.

13. Conclusion

This project demonstrates a complete analytical workflow, starting from raw transactional data and progressing through cleaning, feature engineering, SQL business querying, and dashboard visualization. The findings reveal a business with strong sales performance, valuable repeat customers, and clear product and seasonal patterns. At the same time, the analysis also highlights concentration risk across both customers and geography.